Sample Project: Number Guessing Game

A beginner-friendly Python project inspired by the programming fundamentals in the linked tutorial.

Project overview

The Number Guessing Game asks the player to guess a randomly chosen number within a range. After each guess, the program tells the player whether the guess is too high or too low, and counts the number of attempts.

Main features

• Random number selection
• User input and validation
• Hints for higher or lower guesses
• Attempt counter
• Replay option

Sample code (save as guessing_game.py)

```
import random

def play_game():
    secret_number = random.randint(1, 20)
    attempts = 0

    print("\nWelcome to the Number Guessing Game!")
    print("I'm thinking of a number from 1 to 20.")

    while True:
        entry = input("Enter your guess (or q to quit): ").strip()

        if entry.lower() == "q":
            print("Thanks for playing!")
            return

        if not entry.isdigit():
            print("Please enter a whole number.")
            continue

        guess = int(entry)
        if guess < 1 or guess > 20:
            print("Choose a number between 1 and 20.")
            continue

        attempts += 1

        if guess < secret_number:
            print("Too low. Try a higher number.")
        elif guess > secret_number:
            print("Too high. Try a lower number.")
        else:
            print(f"Correct! You guessed it in {attempts} attempt(s).")
            break

    again = input("Play again? (y/n): ").strip().lower()
    if again == "y":
        play_game()
    else:
        print("Goodbye!")

if __name__ == "__main__":
    play_game()
```

How to run

1. Install Python if it is not already installed.
2. Save the code as **guessing_game.py**.
3. Open a terminal in that folder and run **python guessing_game.py** (on some systems use **python3 guessing_game.py**).
4. Follow the on-screen prompts to play.

Expected result

The program selects a secret number and gives higher/lower hints until the correct guess is entered. It also tracks attempts and offers a replay.

Reference tutorial

Open the supplied YouTube tutorial

Note: The exact tutorial title could not be confirmed from search results; this is a sample beginner project, not a claim that the video teaches this exact project.